



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-2024
PREBOARD EXAMINATION

CLASS: X
SUBJECT: BIOLOGY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this paper is the time allowed for writing the answers.*

*Section A is compulsory. Attempt any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of 7 printed pages.*

SECTION A (40 Marks)
(Attempt all questions from this section)

Question 1

Select the correct answers to the questions from the given options. (Do not copy the question. Write the correct answer only): [15]

- i. Assertion (A): Rapid increase of population was observed in the twentieth century.
Reason (R): Discovery of medicines and vaccination was one of the causes of population increase.
 - a. Both A and R are false
 - b. A is true and R is false
 - c. A is false and R is true
 - d. Both A and R are true
- ii. The movement of plant parts in response to moisture:
 - a. Geotropism
 - b. Hydrotropism
 - c. Thigmotropism
 - d. Chemotropism
- iii. A process by which ovum is released from the ovary:
 - a. Gametogenesis
 - b. Oogenesis
 - c. Spermatogenesis
 - d. Ovulation
- iv. Which part of the eye has the light sensitive rods and cones?
 - a. Conjunctiva
 - b. Cornea
 - c. Choroid
 - d. Retina

- v. The number of live births per 1000 of population per year:
 a. Natality
 b. Mortality
 c. Population
 d. Population density
- vi. Rima was ironing her clothes and accidentally touched the hot iron. Immediately after coming in contact with the hot iron she removed her hand. Which of the following is/are involved in the process described above?
1. Spinal cord
 2. Autonomic Nervous System
 3. Stimulus
 4. Muscle and glands.
- a. Only 1
 b. Both 1 and 3
 c. Only 1, 3 and 4
 d. All 1, 2, 3 and 4
- vii. BS emission norms which were enforced for the entire country on April 1st 2017:
 a. BS I
 b. BS II
 c. BS III
 d. BS IV
- viii. The pigment present in cone cells:
 a. Iodopsin
 b. Rhodopsin
 c. Haemoglobin
 d. Visual Purple
- ix. Assertion (A): Cytokinin helps in cell elongation.
 Reason (R): Cytokinin helps in both cell division and expansion of cotyledons.
 a. Both A and R are true
 b. A is true and R is false
 c. A is false and R is true
 d. Both A and R are false
- x. Phenomenon by which the living or dead plant cells absorb water by surface attraction:
 a. Photosynthesis
 b. Osmosis
 c. Imbibition
 d. Diffusion
- xi. The pigment present in urine:
 a. Glucose
 b. Bile
 c. Urochrome
 d. Protein
- xii. Swachh Bharat Abhiyan was officially launched on
 a. 2nd October 2016
 b. 2nd October 2014
 c. 2nd October 2015
 d. 2nd October 2013
- xiii. The instrument which is used to measure blood pressure:
 a. Sphygmomanometer
 b. Thermometer
 c. Manometer
 d. Potometer

xiv. Shagun went to National Indian Museum and she was observing the evolution lineage of man from ape like ancestors. She understood that man has evolved from ape like ancestors. What are the major changes one will observe going from an ape like form to human form?

1. Reduced cranial capacity
2. Development of chin
3. Reduction of size of canine
4. Reduction of height

- | | |
|--------------------|----------------------|
| a. Only 2 | b. Both 2 and 3 |
| c. Only 1, 3 and 4 | d. All 1, 2, 3 and 4 |

xv. A process by which White Blood Cells squeeze through the walls of the capillaries into the tissues:

- | | |
|---------------|-------------------|
| a. Diapedesis | b. Phagocytosis |
| c. Engulf | d. Transformation |

Question 2

(i) Name the following: [5]

- a. The full form of LH.
- b. A condition in which the lens of the eye turns opaque.
- c. The part of the ear which helps in dynamic balance.
- d. An instrument used to measure the rate of transpiration.
- e. A chemical substance which initiates the process of clotting of blood.

(ii) Choose the odd one out and write the correct category to which the others belong: [5]

- a. Carbon Dioxide, Plastic, Sulphur Dioxide, Carbon Monoxide.
- b. Neutrophil, Monocyte, Eosinophil, Basophil.
- c. Glucose, Starch, Calcium, Cellulose.
- d. Axon, Dendron, Photon, Cyton.
- e. Utricle, Sacculus, Tympanum, Cochlea.

(iii) Fill in the blanks with suitable words: [5]

____a____ is the liquid waste from the domestic activities. Too frequent exposure to ____b____ in a medical diagnostic technique may damage chromosomes in human body. To reduce plastic pollution we must observe the three R's- ____c____, ____d____, ____e____.

(iv) Arrange and rewrite the terms in each group in the correct order so as to be in a logical sequence beginning with the term that is underlined. [5]

- a. Ovulation, Gestation, Fertilization, Child Birth.
- b. Effector, Brain, Stimulus, Receptor.
- c. Chromosome, DNA, Chromatin Fibre, Nucleosomes.

- d. Upper epidermis, Substomatal space, Stoma, Spongy cells.
- e. Conjunctiva, Retina, Optic Nerve, Cornea.

(v) Match the correct answers of Column I with Column II. Please rewrite the correct pairs: [5]

Column I

- a. Cranial Nerves
- b. Eight molecules
- c. Symbol of family welfare
- d. Stroma
- e. Ethylene

Column II

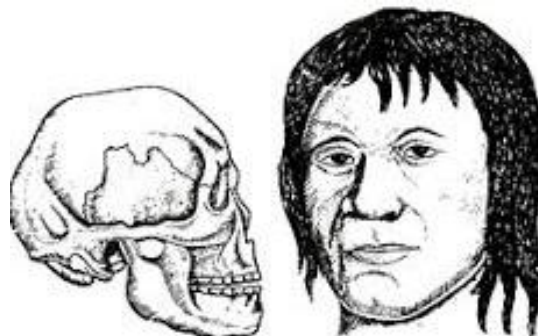
- a. Histone Octamer
- b. Light Independent reaction
- c. Fruits without fertilization
- d. Phytohormone
- e. 12 pairs
- f. Inverted red triangle
- g. 31 pairs

SECTION B (40 Marks)

(Attempt any four questions from this section)

Question 3

- (i) Define Demography. [1]
- (ii) "Testes are present in the scrotum sac outside human male body". Explain why. Mention the function of Leydig Cells. [2]
- (iii) Mention the changes observed in human eye during the dark adaptation. [2]
- (iv) Observe the diagram given below and mention the characteristic features based on cranial capacity and height. [2]



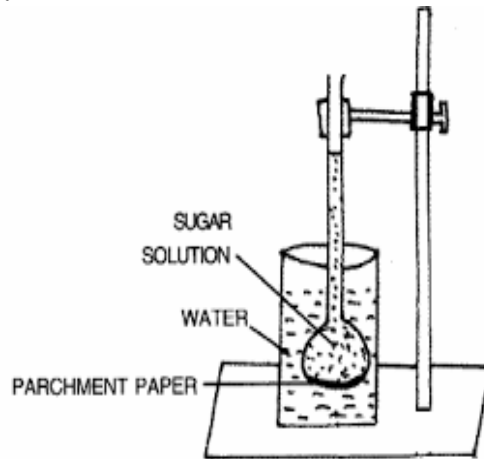
- (v) Draw a neat labelled diagram to explain the correction of Hyperopia. [3]

Question 4

- (i) Define spermatogenesis. [1]
- (ii) What causes Ultrafiltration in Glomerulus? Mention the name of the hormone whose deficiency causes Diabetes insipidus. [2]
- (iii) Mention two important characteristic features of Prophase. [2]
- (iv) Explain why generally male child suffers from colour blindness more than female child. [2]
- (v) Draw a well labelled diagram of Metaphase with a cell having 4 chromosomes. [3]

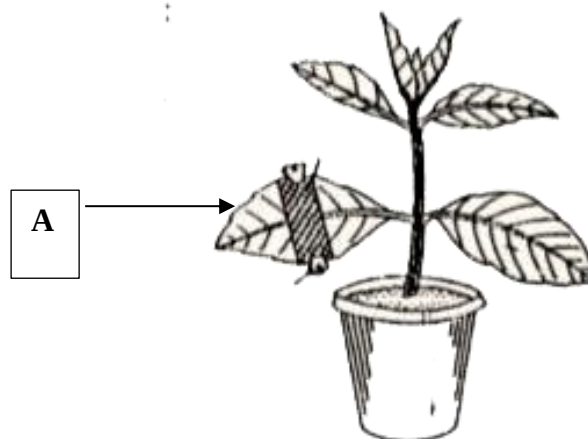
Question 5

- (i) Define the term flaccidity. [1]
- (ii) Answer the following questions: [2]
 - a. Observe the diagram given below and explain the process shown in the diagram.



- b. Why do grass in the lawn die when we sprinkle salt on them?

- (iii) Mention two reasons for population explosion in India. [2]
- (iv) Differentiate between Karyotype and Karyokinesis (definition) [2]
- (v) Observe the diagram given below and answer the following questions: [3]



- a. Mention the aim of the experiment.
 - b. What do you observe in leaf "A" at the end of the experiment when tested with Iodine solution?

Question 6

- (i) Mention the scientific name of peppered moth. [1]
- (ii) Mention two functions of Cerebellum. [2]
- (iii) State the Law of Segregation. Mention the scientific name for the plant used by Mendel in his experiment. [2]

- (iv) A pure plant with axillary flower (AA) is crossed with a pure plant with terminal flower (aa). Find out the Gametes and Genotypic ratio of F₂ generation. [2]

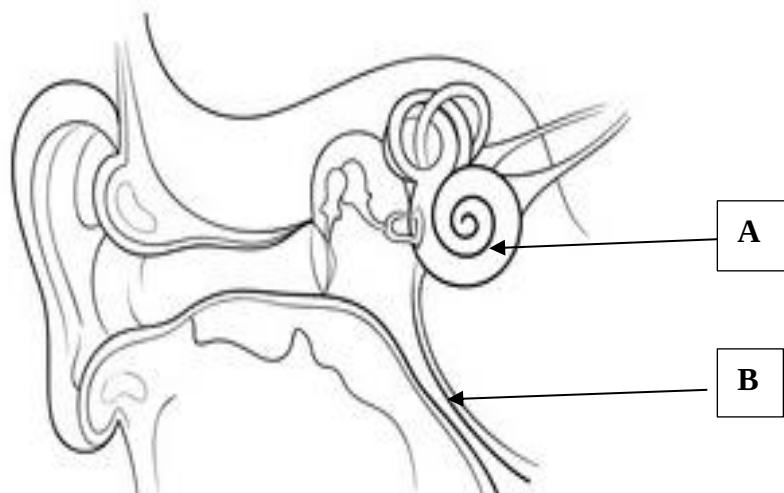
- (v) Answer the following questions by observing the diagram given below. [3]



- Identify and define the pollution shown in the above diagram.
- Mention one important source of the above pollution.
- Mention one harmful effect of the above pollution on birds.

Question 7

- Name the instrument which helps to demonstrate geotropism. [1]
- Define synapse. Mention the function of corpus callosum. [2]
- Explain the term gestation. How long does gestation last in human body? [2]
- Observe the diagram of Human ear given below and mention the function of the parts marked as “A” and “B”. [2]



- (v) Draw a neat, labelled diagram of a gamete which contains only X chromosome. [3]

Question 8

- Explain why AB is known as Universal recipient. [1]
- Mention any two objectives of Swachh Bharat Abhiyan. [2]
- Mention the location of seminal vesicle and its function. [2]

- (iv) **Mention the two functions of Vitreous Humour.** [2]
- (v) **Observe the male reproductive system shown in the diagram given below and label 'A', 'B'. Mention the function of the part labelled as 'C'.** [3]

